

Pre-class preparation

Please read the following textbook sections from Degroot and Schervish's *Probability and Statistics* (fourth edition) or watch the video, as indicated:

- Video: <https://expl.ai/ANABNQT>

Objectives

By the end of the day's class, students should be able to do the following:

- Explain the equivalency between confidence sets and hypothesis tests.
- Demonstrate for particular models how to create a confidence interval from a hypothesis test, and vice versa.

Reflection Questions

Please submit your answers to the following questions to the corresponding Canvas assignment by 9:00AM:

1. Suppose $X_1, \dots, X_{25} | \mu \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ where μ is unknown and σ^2 known. We have previously shown that a $\gamma \times 100\%$ CI for μ is $(\bar{X} - \Phi^{-1}(\frac{1+\gamma}{2}) \frac{\sigma}{\sqrt{n}}, \bar{X} + \Phi^{-1}(\frac{1+\gamma}{2}) \frac{\sigma}{\sqrt{n}})$. If I'd like to test the hypothesis $H_0 : \mu = \mu_0$ at the level-0.10 level, how would I use the confidence interval to do so?
2. (Optional) Is there anything from the pre-class preparation that you have questions about? What topics would you like some more clarification on?